# DA21 Build and check the chart report

Week 7, Show it | Session 21 of 24 | Python for Data Analysis

**Today's question.** Does my report answer my questions without a verbal tour?

**Objective.** By the end: you can build a chart report that a stranger reads in 30 seconds, and you can critique a coding assistant's hint with a test.

> **WHERE THIS IS USED**
> A report that needs its author in the room has not done its job. Clear titles, labels and units let a farmer, a manager or a parent read it alone. Could your report speak for itself?

Before this session: DA20 Build a reliable analysis. If you cannot explain the last example, use its error card with the instructor first.

## Words for today

| Word | What it means | Think of it like this |
|---|---|---|
| chart report | A short notebook of charts, numbers and sentences that answers the project questions. | A news page with pictures and captions. |
| caption | One sentence under a chart that says what it shows. | The words under a photograph. |
| data source note | A line that says where the data came from and its limits. | The credit under a map. |

## Make

Build the chart report. Use few charts, readable labels with units, the covered dates and a visible data source note. Write two findings and one limitation next to their evidence. Give a peer 30 seconds to find the largest category and its total, and record where they hesitate. To put a chart in your report, call plt.savefig("revenue.png", bbox_inches="tight") before plt.show(), then insert the PNG with Insert and Pictures. When the instructor confirms you are ready, spend 20 minutes with AI_CODING_ASSISTANTS_LAB.pdf: critique a hint on the practice function, run a test and log it in AI_USE_LOG.md.

Save your work as DA21. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

## Error card

| You see | It means | Fix |
|---|---|---|
| The saved picture cuts off the labels | The figure was saved without room for the labels. | Add bbox_inches="tight" to plt.savefig. |
| The picture is saved blank | plt.show() ran before plt.savefig, which clears the figure. | Call plt.savefig before plt.show. |

## Exit ticket

A reader can find my main finding in 30 seconds, and I tested the assistant's advice before I used it.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.